Irving Cristóbal Guerra Vargas

Software Engineer

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Software Engineer with 8+ years architecting and shipping production systems end to end: mobile, web, backend, desktop and IoT. Lead engineer of the FTN app at HealthyVita and builder of live platforms for companies across Mexico. I engineer with AI agents as a disciplined part of the process (layered context, rules, MCP, subagents and production verification), with ~2,850 commits co-authored with Claude, and I build AI features into products: chatbots, operations-analysis agents and neural text-to-speech.

EXPERIENCE

HealthyVita · FTN · Software Engineer — 2024 – present

- Lead engineer of the FTN subscription fitness app (Flutter, iOS & Android): 1,599 of its 1,753 commits.
- Full-stack across the Django REST API (Postgres) and the Angular backoffice and public web, in a team of ~15.
- In-app purchases plus 4 payment channels, Apple Health / Health Connect, Shorebird code push, AWS (ECS, Lambda, Terraform).
- Integrated AI features and modeled LLM cost per conversation to choose the right model per task.

MDreieck · Nuntius · Lead engineer (via Antware, independent since 2024) — 2020 – present

- Early-warning platform for Mexico: seismic alert reception, CAP alerts by territory, SAME/EAS radio rebroadcast, IoT siren fleet, weather and river stations with automatic thresholds.
- Real-time control panel (Meteor, Vue, MongoDB, WebSockets), Flutter app with full-screen critical alerts on App Store and Google Play.
- Three Electron/Next.js desktop apps, Raspberry Pi (Node.js) and ESP32 (C++) firmware, neural TTS voices (Piper, Google TTS), Docker server per client.
- ~1,290 commits.

Kabandu · Freelance · Founding engineer — 2025 – present

- Operating system for a brow & beauty salon chain: POS and admin desktop (Electron, fingerprint check-in), bookings, gift cards, loyalty and memberships.
- Client and staff apps in Flutter (App Store, Google Play), TypeScript server on Fly.io, WhatsApp Business automation. ~1,200 commits.

Krispis Breakfast n' Dinner · Freelance · Full-stack engineer — 2023 – present

- Multi-branch restaurant platform: online ordering (Next.js, Stripe), kiosk, table QR, kitchen display, POS, CRM and inventory.
- TypeScript server on Fly.io with Uber Direct delivery, Google Wallet, web push and WhatsApp OTP. ~1,160 commits.

Antware · Software Engineer · Embedded Systems Developer — 2020 – 2024

- React Native, Meteor (MongoDB), Vue and React, real-time WebSockets, Jest, Scrum, GitHub CI.
- Seismic-alert systems with Raspberry Pi and Node.js, EAS-SAME desktop transmitters, compliance with Mexican seismic-alert standards.

Carklin · Full-stack Developer — 2021 – 2022

- React Native app, REST APIs with Cloud Functions, push notifications, in-app purchases with RevenueCat.

IPN · Full-stack Developer — 2019 – 2020

- React and Firebase, PHP and MySQL, Java (Spring Boot, Struts 2, JSF, Hibernate), React Native.

Freelance · 10+ projects 2018 – 2024

- Bongless (restaurant group), Relpers, Neurona Wireless (real-time voice requests with Dialogflow and Google Assistant), MyBike (App Store and Google Play).

OWN PRODUCTS

LiteDash: open-source Android car launcher + companion app (Kotlin, Compose; Google Play, litedash.dev) · LGlow: open-source ad-free home screen for rooted LG TVs (webOS) · Duo Arena: online platform fighter, 2–6 players (Flutter, Flame; duoarena.com.mx) · Derrapazo: 3D kart racer (Godot; Google Play) · Get My Currency (Google Play).

SKILLS

AI engineering: Claude Code, agentic workflows, MCP, subagents, LLM cost modeling, TTS

Mobile: Flutter, Dart, Kotlin, Jetpack Compose, React Native, Shorebird

Web: TypeScript, React, Next.js, Angular, Vue, MUI

Backend: Node.js, Express, Python, Django, Meteor, Java Spring

Data & cloud: PostgreSQL, MongoDB, Firebase, AWS, Fly.io, Terraform, Docker

Other: Electron, Godot, Raspberry Pi, ESP32

EDUCATION

B.S. Computer Systems Engineering
ESCOM, Instituto Politécnico Nacional · 2015 – 2020

Digital Systems Technician
CECyT 1, Instituto Politécnico Nacional · 2013 – 2015